



Introduction to Python



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VS Code Setup

VS Code is a code editor used to write, organize, and run source-code files. For Python practice, students can create a .py file, write Python code, and run it using the Python environment installed on the system.

Basic setup:

- Install Python
- Install VS Code
- Install the Python extension
- Create a file such as main.py
- Run a simple print() statement

What is Python ?

Python is an interpreted, high-level, and general-purpose programming language.

Why learn Python ?

- Simple and beginner-friendly syntax
- Easy to read and write code
- Works on many platforms (Windows, Mac, Linux)

Python and Its Language Level

- Python is a high-level programming language.
 - This means you can write code in simple, English-like words.
 - You don't have to worry about how memory or hardware works in the background.
- High-Level Language
 - Easy to read and write.
 - Handles memory and hardware automatically.
such as: Python, Java.
- Mid-Level Language
 - Has features of both high-level and low-level.
 - Gives more control but still readable. such as: C, C++.
- Low-Level Language
 - Very close to machine code (0s and 1s).
 - Directly interacts with hardware, very fast but hard to understand and write. such as: Assembly, Machine Language.

Example:

```
print("Hello, Python!")
```

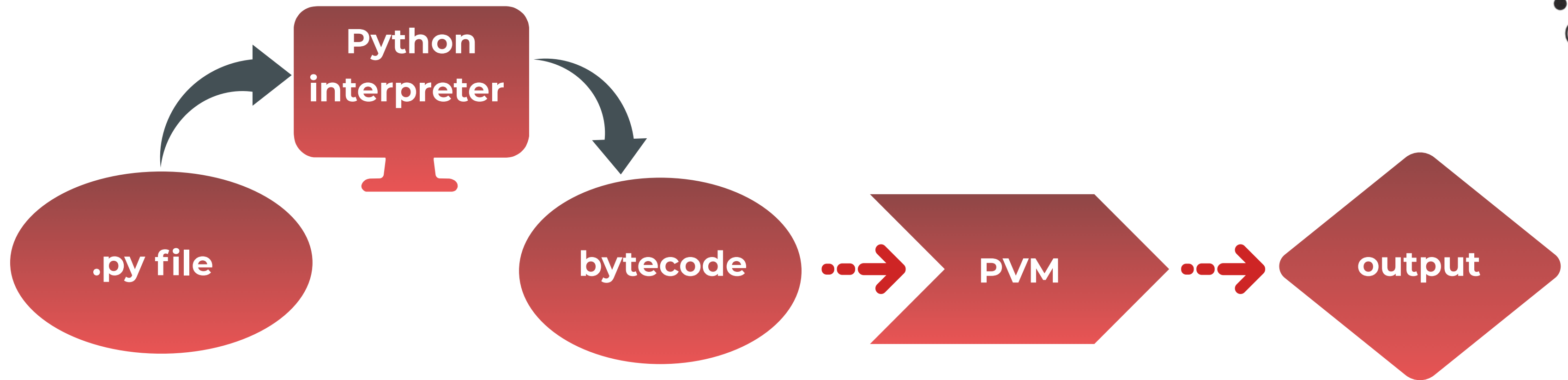
Output: Hello, Python!

Difference between Interpreter and compiler.

Feature	Interpreter	Compiler
1.Execution Style	1.Execute code line by line	1.Translates entire code at once
2.Speed	2.Slower (executes every line separately)	2.Faster (runs machine code directly)
3.Error Handling	3.Shows errors one at a time	3.Shows all errors together
4.Example languages	4.Python, JavaScript, Ruby	4.C, C++, Java

- Python is called an interpreted language because its code is executed line-by-line by the Python interpreter.
- It does not need to be compiled into machine code before running – instead, it is directly executed by the interpreter.
- **Python uses an interpreter, which makes development easier, especially for beginners**

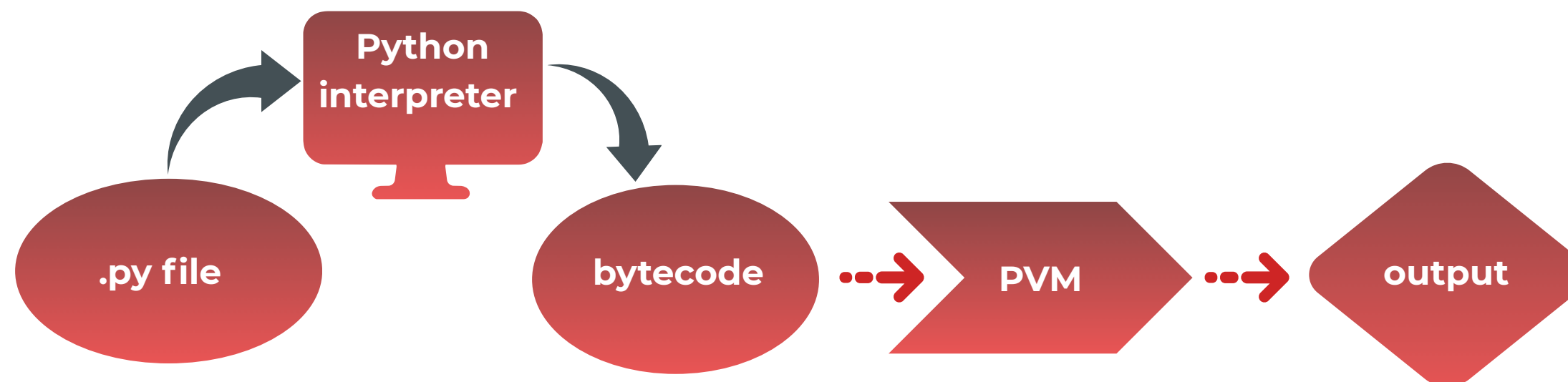
Python as an Interpreted Language :



Execution Flow

Python code commonly follows this beginner-level model.

1. Write source code in a .py file.
2. Python processes the source code.
3. Python commonly produces bytecode.
4. The Python Virtual Machine (PVM) executes that bytecode.
5. The program produces output.



Variables

Variables are used to store data.

Python is dynamically typed - you don't need to declare the type.

Example: `x = 5`
`name = "joker"`

Naming Rules

- Must start with a letter or underscore (_).
- Can contain letters, digits, and underscores.
- Case-sensitive (name and Name are different).
- Cannot use Python keywords like- for, while, if, etc.

camelCase: First word lowercase, subsequent words capitalized.

Example:

`firstName,`
`iAmVeryTiredToday`

PascalCase: Every word starts with a capital letter.

Example:

`MacBook,`
`ThisIsAGoodIdea`

snake_case: All lowercase with underscores between words.

Example:

`first_name,`
`i_just_want_a_nap`

Input & Output

Input : Input is the process of receiving data from the user.

Example:

```
name = input("Enter your name: ")  
age = int(input("Enter your age: "))  
height = float(input("Enter height: "))
```

Output : Output is information produced by a program and displayed to the user.

Example:

```
print("Hello", name)  
print("Hello")
```

Comments

Single-line :

```
# This is a single-line comment
```

Multi-line :

```
"""This is a multi-line comment"""
```

Summary

- **Introduction to python:** python is a simple, high level, and versatile programming language used for many types of applications.
- **Python as Interpreted Language:** Python executes code line by line using an interpreter instead of compiling the entire program first.
- **Execution flow:** Execution flow is the order in which the statements of a Python program are executed.
- **Variables:** Variables are the names used to store and refer to data values in a program.
- **Input and Output:** `input( )` is used to take data from user, while `print( )` is used to display outputs.

Next Steps:

- Python Data Types & Operators
- Numbers, Strings, and Booleans
- Basic Arithmetic & Comparison Operators
- Type Conversion and Casting
- Introduction to Logical Operators

*Thank
You*

